Splash Rules II: quasicrystal approximants

Every count of the periodic approximants of the Fibonacci chain, the Ammann–Beenker tiling and the 3D Penrose tiling, exactly, and the one number that orders them



DICTO · study of 31 October 2026 · results for DICTO, not for publication

DICTO's idea (Splash Rules, 27 October): plot a whole family of exact values and look for one rule that covers the local ones. Third outside run, after the perovskites and the Frank–Kasper phases, aimed at DICTO's long-term aim: a bridge between periodic and aperiodic order. Computation with Claude (Anthropic).

Verdict. Yes, one number orders the approximants: the share of one tile (S tiles in the Fibonacci chain, squares in the Ammann–Beenker tiling, prolate rhombohedra in 3D). Every count we computed is a straight line in it on one stretch of approximants, the *chamber*, that contains the quasicrystal, and the quasicrystal lies exactly on every one of those lines. Pair counts and vertex counts (window 2) need no second number there: the approximants are the words and tilings in which the Swap defect term of the Window Law is pinned at zero. The defect appears only beyond the chamber walls, as a kink: in 1D the kink at a = 1/2 is exactly twice the Swap defect; in 2D a forbidden vertex star appears whose number grows as a square, $2(3a - 2b)^2$ or $(4a - 3b)^2$, and every other count shifts by a fixed vector of it. So the defect term does not shrink towards the limit; it is already zero for every approximant inside the chamber, and what shrinks is only the distance along the line (the phason strain).

Known and new. The 1D part is classical (Sturmian words: Morse–Hedlund, Berthé 1996, the three-distance theorem); the vertex counts of the Pell-number square approximants are tabulated by Jagannathan and Duneau (2023). Not found and, as far as we checked, new: that all 42 square approximants with $2/3 \le t \le 3/4$ (not only the Pell ones) obey six fixed linear laws in the numbers of rhombs and squares, that 2/3 and 3/4 are walls, and the square-law defect with its transfer vector outside them; the cubic-cell tile formulas of the 3D approximants, prolate $4(p^3 + 3p^2q + q^3)$ and oblate $4(p^3 + 3pq^2 - q^3)$.

1. Set-up

All three families are cut-and-project sets: lattice points of Z^n are kept when they lie in a strip around a plane E, and projected. The quasicrystal takes an irrational E (slope $1/\varphi^2$, angle with $\sqrt{2}$, or icosahedral with φ); an approximant replaces the irrational number by a fraction, E becomes rational, and the pattern repeats. We used the canonical strip (the plane plus a unit cube) and a generic offset, and counted everything per period, exactly:

· **1D** (scripts/sturm.py): the word $s_n = \lfloor (n+1)a + r \rfloor - \lfloor na + r \rfloor$, letter S (short) for 1 and L (long) for 0, a = share of S. For a = p/q the word repeats after q letters. All 1101 fractions p/q with $q \le 60$, every window (factor) length g = 1 to 9, the Fibonacci chain $a^* = 1/\varphi^2 = (3 - \sqrt{5})/2$ in exact arithmetic in $Q(\sqrt{5})$, and for contrast all 747 primitive periodic words of length up to 12.

· **2D** (scripts/ab.py, abfam.py, ablaw.py): Z^4 to the plane, star e_k at angles $k\pi/4$, Ammann–

Beenker (AB) tiles: squares (edges e_0, e_2 or e_1, e_3) and 45° rhombs. A rational plane E′ with period lattice $\Lambda = Z^4 \cap E'$ is handled by an integer quotient map $Z^4 \to Z^2$ with kernel Λ: one period of vertices is the set of integer points in a zonogon with integer edges, so every count is an integer and every acceptance area a fraction. Square approximants: √2/2 replaced by t = a/b in the rows (1, t, 0, −t), (0, t, 1, t); all 246 fractions $1/2 \le t \le 1$ with $b \le 40$, plus the Pell ones up to 99/140. For contrast, 240 random rational planes of the same orientation. The quasicrystal in Q(√2).

· **3D** (scripts/ico.py): Z^6 to space with the icosahedral star (±1, φ, 0) and its cyclic shifts; cubic approximants replace φ by F_{k+1}/F_k = 1/1, 2/1, 3/2, … 55/34; tile counts only.

No floats enter any test; the splash runs on dicto-big-brother in seconds. Two facts used throughout and checked on every run: the number of tiles with edges {i, j} (or {i, j, k}) per period is the absolute 2×2 (3×3) minor of a basis of Λ — its Plücker coordinate — and the number of vertices per period equals the number of tiles.

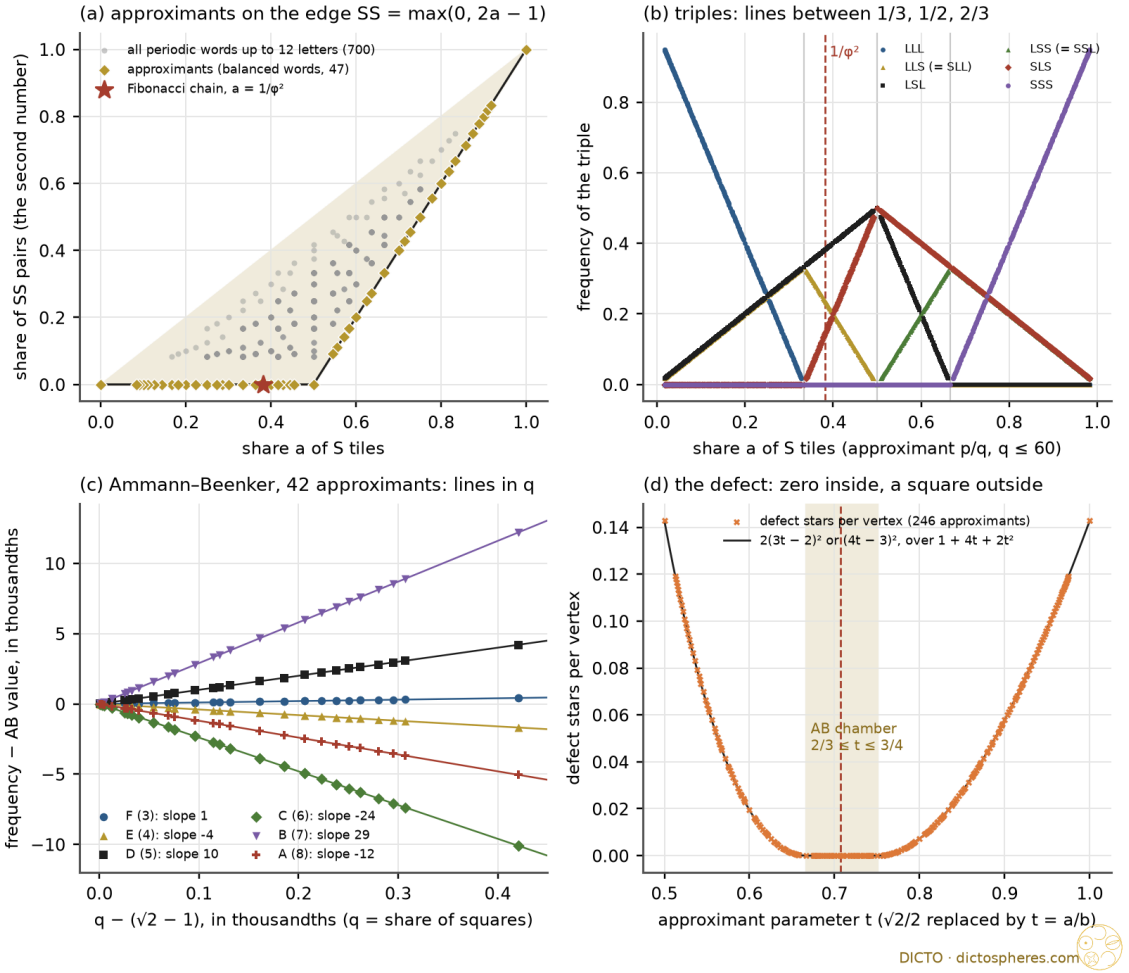


Figure 1. The splash. (a) Window 2 in 1D: share of SS pairs against the share a of S tiles for every periodic word up to 12 letters (grey) and the approximants (gold); the approximants form the lower edge max(0, 2a − 1), the Fibonacci chain sits at (1/φ², 0). (b) Window 3: the frequency of every triple in every approximant p/q, q ≤ 60, lies on straight lines that bend only at 1/3, 1/2 and 2/3. (c) Ammann–Beenker square approximants inside the chamber 2/3 ≤ t ≤ 3/4: each vertex-star frequency minus its AB value, against the square share minus √2 − 1; six straight lines through the origin. (d) The forbidden star (two squares side by side at a vertex): none inside the chamber, a square law outside.

2. The Fibonacci chain and its approximants

> **Theorem 1 (Farey Window Law; classical in substance).** For every window length g and every factor w of length g, the frequency of w in the approximant a = p/q is a straight line in a on each interval between neighbouring fractions with denominator ≤ g, and continuous across them. On the interval containing $1/\varphi^2$ the Fibonacci chain lies on the same line, with values in $\mathbb{Q}(\sqrt{5})$; the approximant F_{n-2}/F_n lies on the chain's window-g line exactly when $g < F_{n+2}$, and there its frequency differs from the chain's by (slope) × $(p/q - 1/\varphi^2)$.

Proof. The factor of length g starting at a letter is fixed by where nα + r falls among the g + 1 points −ka mod 1 (k = 0 … g); its frequency is the length of its interval, a difference of two such points, i.e. an integer combination m − ka, a line in a. Two points meet only when ka is a whole number, at fractions with denominator ≤ g; there the order changes and the line bends. For a = p/q the points lie on the grid of q-ths and a generic offset makes the count per period equal to q times the length exactly. The rule for F_{n-2}/F_n is the closed Farey interval of order g around $1/\varphi^2$ (checked for g ≤ 300). ∎ Checked on all 1101 approximants for g ≤ 9: every frequency on every Farey interval is one line; complexity min(g + 1, q) factors; at most three distinct frequencies per length (the three-distance theorem, Berthé 1996); every approximant balanced.

The Window Law in 1D. At window 2 the perovskite form reads F(LL) + [F(LS) + F(SL) − 2F(LL)]·a + Δ·p_{SS} with Δ = F(LL) + F(SS) − F(LS) − F(SL), the Swap defect. For approximants the second number is not free: p_{SS} = max(0, 2a − 1), the smallest share of SS pairs any word with this a can have (Figure 1a; the other 700 words lie above). So every pair count of an approximant is a line in a on each side of 1/2, and its kink at 1/2 is exactly 2Δ: the Swap Rule (Δ = 0) holds exactly when the kink vanishes. On the Fibonacci side (a < 1/2), p_{SS} = 0 for every approximant, not just in the limit: SS never occurs, as in #69. The kinks at longer windows follow the same pattern with the denominator as factor (window 3: at 1/3 the vector +3, −3, −3, +3 on LLL, LLS, SLL, SLS; at 1/2 the vector +2, −4, +2, +2, −4, +2 on LLS, LSL, LSS, SLL, SLS, SSL).

window	chamber of $1/\varphi^2$	factor frequencies on it (lines in a), value at a = $1/\varphi^2$	approximants on the line
1	[0, 1]	S: a = (3 − √5)/2; L: 1 − a	all
2	[0, 1/2]	LS = SL: a; LL: 1 − 2a = √5 − 2; SS: 0	from 1/2
3	[1/3, 1/2]	LLS = SLL: 1 − 2a = √5 − 2; LSL: a; SLS: 3a − 1 = (7 − 3√5)/2	from 1/2
5	[1/3, 2/5]	adds LLSLL: 2 − 5a = (5√5 − 11)/2	from 1/3
8, 9	[3/8, 2/5]	three values: 2 − 5a, 3a − 1, 8a − 3 = 9 − 4√5	from 2/5

Status: classical (Morse and Hedlund 1940 for balanced words and complexity; Berthé 1996 for the frequencies; the three-distance theorem). Ours: reading the Swap defect as half the kink, and the approximants as the lower edge of the Window-Law plane.

3. The Ammann–Beenker tiling and its square approximants

> **Result 2** (exact on all 42 square approximants with $2/3 \le t \le 3/4$, $b \le 40$, and on the Pell ones up to 70/99 and 99/140). With R rhombs and Q squares per period, the six vertex stars of the AB tiling occur
>
> $$F\ (3\ \text{edges}) = Q,\ E\ (4) = 2R - 2Q,\ D\ (5) = -4R + 6Q,\ C\ (6) = 10R - 14Q,\ B\ (7) = -12R + 17Q,\ A\ (8) = 5R - 7Q,$$
>
> and no other star occurs. Per vertex these are straight lines in the square share $q = Q/(R + Q)$; at $q = \sqrt{2} - 1$ they give exactly the AB frequencies $\sqrt{2} - 1$, $6 - 4\sqrt{2}$, $10\sqrt{2} - 14$, $34 - 24\sqrt{2}$, $29\sqrt{2} - 41$, $17 - 12\sqrt{2}$ ($= 1/\lambda, 2/\lambda^2, 2/\lambda^3, 2/\lambda^4, 1/\lambda^5, 1/\lambda^4$ with $\lambda = 1 + \sqrt{2}$). Edges counted by the stars at their two ends (9 kinds) are lines in (R, Q) on the same chamber, e.g. F–E edges $4R - 4Q$, D–D edges $-2R + 3Q$.

So in 2D the share of squares plays the part of the hexagonality: one number, every count a line, the quasicrystal on every line (Figure 1c). The counts do not depend on the offset of the strip: for every square approximant the integer counts equal the exact areas of the acceptance domains. Example $t = 5/7$: 239 vertices per period, $R = 140$, $Q = 99$, stars A 7, B 3, C 14, D 34, E 82, F 99 (Figure 2).

The walls and the defect. Outside the chamber a seventh star appears: six edges with the two squares side by side at the vertex (AB never has this). Its number per cell of $N_u = b^2 + 4ab + 2a^2$ vertices is

$$X = 2(3a - 2b)^2 \text{ for } t = a/b < 2/3, \quad X = (4a - 3b)^2 \text{ for } t > 3/4, \quad X = 0 \text{ inside},$$

and every count is the chamber line plus a fixed multiple of X: (F, E, D, C, B, A, defect) shift by $(0, 0, -1, +2, -3, +1, +1)\cdot X$ (checked on all 246 square approximants, 1722 counts). This is the Window-Law shape again: a line in one number plus a defect term with a fixed coefficient vector, the 2D counterpart of the Swap defect +1, −2, +1. The new feature is the square: in 1D the forbidden SS pairs grow linearly beyond the wall ($2a - 1$), in 2D the forbidden star grows quadratically, because its acceptance domain is born at a point and grows in two directions (Figure 1d). The walls are the Pell approximant $t = 2/3$ and its quarter-turned partner $3/4 = 1/(2\cdot 2/3)$; 1/2 and 1 (the first Pell approximant and its partner) lie outside.

General approximants. For a rational plane without the four-fold symmetry the vertex counts are not fixed by the plane: on 229 of the 240 random planes (three offsets each), moving the offset of the strip changes them (a row of phason flips), e.g. one plane with tile counts 15, 30, 20, 27, 31, 26 gives 64 or 63 three-edge stars. Only the averages over the offset are fixed, and those are areas with large denominators, not linear in the tile counts. The tile counts themselves are always the Plücker coordinates. So the one-number law in 2D belongs to the symmetric (square) approximants, the ones used in practice.

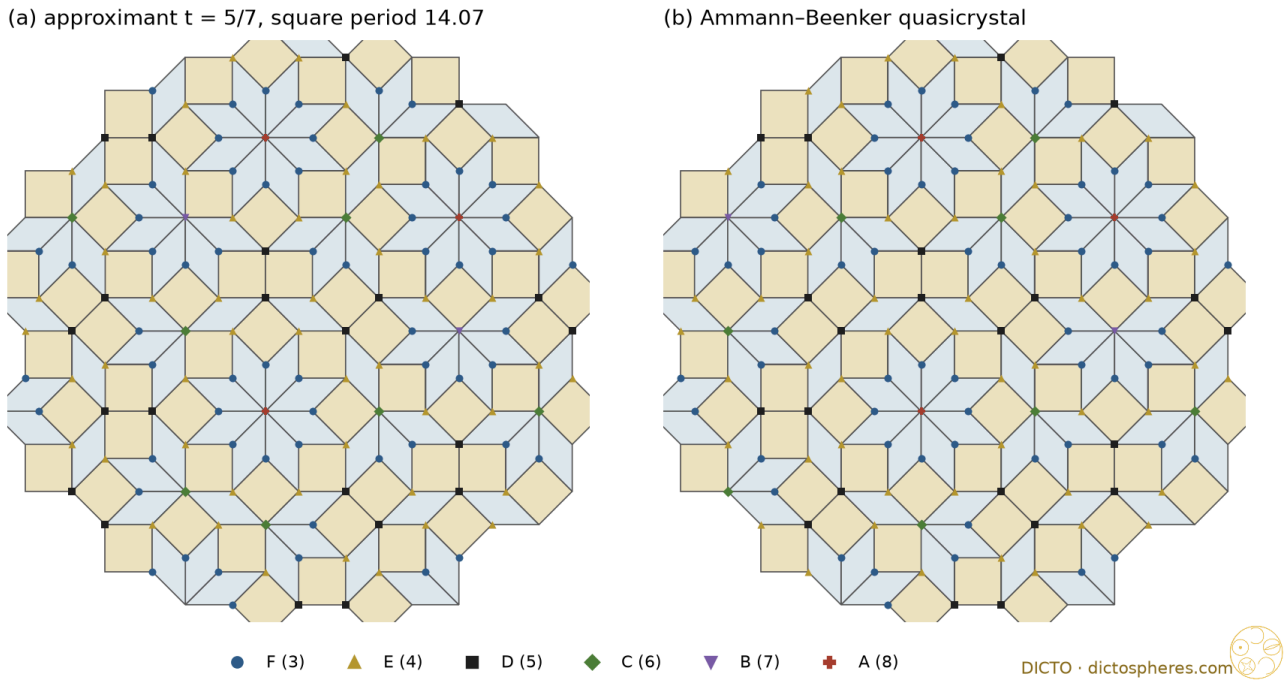


Figure 2. (a) The square approximant t = 5/7 (239 vertices per period, the period a square of side 14.07 edges, larger than the patch) and (b) the Ammann–Beenker quasicrystal, both drawn with the AB rhombs (blue) and squares (gold), vertices marked by their star. Around the centre the two agree tile for tile; they differ only along a few rows of flips.

4. The 3D Penrose tiling and its cubic approximants (tile counts)

> **Result 3** (exact, 1/1 to 55/34). For the cubic approximant $\varphi \approx p/q$, the cubic cell spanned by the rows of the rational projection (scaled by q) holds
>
> $$\text{prolate} = 4(p^3 + 3p^2q + q^3), \quad \text{oblate} = 4(p^3 + 3pq^2 - q^3)$$
>
> rhombohedra, and as many vertices as rhombohedra. At $p/q = \varphi$ the ratio is exactly $\varphi : 1$ (prolate share $1/\varphi$, $Q(\sqrt{5})$).

approximant	1/1	2/1	3/2	5/3	8/5	13/8	21/13
prolate + oblate per cubic cell	20 + 12	84 + 52	356 + 220	1508 + 932	6388 + 3948	27060 + 16724	114628 + 70844
primitive cells per cubic cell	2	1	1	2	1	1	2
prolate share	5/8	21/34	89/144	377/610	1597/2584	6765/10946	28657/46368

Every tile orientation agrees in sign with the quasicrystal's, so these are proper canonical tilings (the 1/1 lacks 4 of the 20 orientations). The prolate share of the approximant F_{k+1}/F_k is itself the Fibonacci ratio F_{3k+2}/F_{3k+3} (all nine computed): the 3D approximants carry the 1D approximants of $1/\varphi$ in their tile counts, three steps at a time. The tile counts are window 1, so they are lines in the prolate share by definition; vertex types (24 kinds, Katz; 39 with decorations, Hann, Socolar and Steinhardt 2016) were not run.

5. What is known, what is new

law	exact on	status
1D: window-g frequencies are lines in a between Farey points of order g; ≤ 3 values; complexity g + 1; approximants balanced	1101 approximants, g ≤ 9	classical (Morse–Hedlund; Berthé 1996; three-distance theorem)
1D: F_{n-2}/F_n on the Fibonacci window-g line iff $g < F_{n+2}$; offset from it = slope × phason strain	g ≤ 300 (Farey), g ≤ 9 (counts)	known in substance; this statement not found
1D: approximants = lower edge $p_{SS} = \max(0, 2a - 1)$ of the Window-Law plane; kink at 1/2 = 2 × Swap defect	747 words	elementary; ours as framing
tile counts per period = Plücker coordinates of the period lattice; vertices = tiles	every plane, every offset	classical (canonical projection)
AB frequencies 1/λ … in Q(√2); square-approximant counts for the Pell sequence	exact	known (Baake–Grimm; Jagannathan–Duneau 2023, Table 1)
AB: six star laws linear in (R, Q) on the whole chamber 2/3 ≤ t ≤ 3/4; edge-pair laws; no other star there	42 approximants	not found; new (empirical, exact)
AB: walls 2/3, 3/4; defect star $X = 2(3a - 2b)^2, (4a - 3b)^2$; transfer vector (0, 0, −1, +2, −3, +1, +1)	246 approximants	not found; new (empirical, exact)
AB: non-square approximants have offset-dependent vertex counts	229 of 240	known in substance (phason flips by window shifts)
3D: prolate $4(p^3 + 3p^2q + q^3)$, oblate $4(p^3 + 3pq^2 - q^3)$ per cubic cell; 1/1 = 20 + 12; prolate share F_{3k+2}/F_{3k+3}	9 approximants	formula not found; minors method classical

6. Prior art: sources checked

Reference library (ask-library, seven searches: Sturmian factor frequencies / three-distance; periodic approximants cut and project; Ammann–Beenker vertex frequencies; Penrose vertex types de Bruijn; phason strain approximant linear; icosahedral cubic approximant 1/1 prolate oblate; Christoffel balanced words). **Read**: Kléman 2013, Defects in quasicrystals revisited I (pp. 1–5: approximants differ from the quasicrystal by flips, one per period for Fibonacci approximants); Baake 1999, A guide to mathematical quasicrystals (§13: defective vertex configurations, Baake–Joseph 1990 cited); Hann, Socolar and Steinhardt 2016, Local growth of icosahedral tilings (p. 3: acceptance domains, 24 vertex geometries, 39 cells). Catalogue entries only: Goldman and Kelton 1993; Yamada et al. 2021 (cubic approximants by linear phason strain); de Bruijn 1981 I, II; Fujita et al. 2013; Kramer 2011; Akiyama et al. 2026 (Sturmian lattices). **Web**: "Berthé 1996 frequencies of factors of Sturmian sequences" (Berthé, Fréquences des facteurs des suites sturmiennes, Theor. Comput. Sci. 165 (1996) 295–309, via bibliographic entry and citing surveys; full text not read); "Ammann-Beenker tiling periodic approximants vertex configurations frequencies square approximant defect" (read: A. Jagannathan and M. Duneau, Properties of the Ammann–Beenker tiling and its square approximants, arXiv 2308.07701 / Israel J. Chem. 2024, §§2.2, 3, 3.1, 3.2: AB frequencies (2.5), Pell approximants, Table 1 of vertex areas for n = 2 … 5, which our counts reproduce, rows of phason flips; search summaries of arXiv 1009.0136 and Mazáč 2212.09406); "1/1 cubic approximant 3D Penrose tiling 20 prolate 12 oblate" and a second query on the same (no source gives tile counts per cell; summaries of arXiv cond-mat/0402370, 2108.03341, cond-mat/0405298). **In-house**: #69 The Hexagonality Law and the Fibonacci Stacking (built on, not repeated: #69 has the letter and pair frequencies of the infinite chain and the stackings 15, 16, 26, 63); Splash Rules II perovskites and Frank–Kasper (Window Law, Swap pattern). **Classical, cited not read**: Morse and Hedlund 1940; Lothaire; Baake and Grimm, Aperiodic Order vol. 1 (2013); Duneau, Mosseri and Oguey 1989; Elser and Henley 1985; Katz 1988.

7. Conclusion and status

Conclusion: the approximants of a quasicrystal are ordered by one number, the share of one tile; on the chamber that holds the quasicrystal every count is a straight line in it with the quasicrystal on the line, because the approximants are exactly the patterns with the Window Law's defect term at zero; the defect lives only beyond the chamber walls, linear in 1D (twice the Swap defect at the kink) and a square in 2D. **Status:** done for 1D and 2D counts and 3D tile counts (study 2026-10-31-splash2-approximants). Not run, decided: Penrose rhombus approximants (the 2D case is covered by AB, whose window is a single octagon) and the 3D vertex types (a 3D acceptance arrangement; queue only if DICTO wants it).